

DICS v2 Data Flow — Overview & How to Read This Set

Why 15 documents instead of 48

The literal request was 8 scenarios × 6 steps = 48 documents. Building that literally would mean re-explaining the same proxy mechanics, the same EIP-712 signing process, and the same payout logic 8 times over — only the dollar amounts and which branch gets taken actually differ between scenarios. That's not more understandable, it's the same explanation with more scrolling.

Instead: **6 Step documents** hold every mechanical detail, written once, in full depth. **8 Scenario documents** are short and concrete — each one states exactly what's different for that case (the numbers, the branch taken, what reverts and why) and points back to the relevant Step document rather than repeating it. Read a Scenario document first for the concrete story; follow its references into the Step documents for the underlying mechanics.

The address book — used identically across every document in this set

Fixed, synthetic addresses so every document traces the *same* values through the *same* places, making it possible to cross-reference between documents without re-deriving anything:

Name	Address	Role
Alice	0x11111111111111111111111111111111AAAA	Policyholder (primary example)
Bob	0x12121212121212121212121212121212BBBB	Policyholder (used in multi-user scenarios)
Attacker	0x99999999999999999999999999999999CCCC	Unauthorized actor (Scenario 8)
InsurancePolicy proxy	0x22222222222222222222222222222222BBBB	Policy catalog + premium collection
ClaimRegistry proxy	0x33333333333333333333333333333333CCCC	Claim lifecycle + payout
OracleAdapter proxy	0x44444444444444444444444444444444DDDD	Signature verification
StableCoin (dUSD)	0x55555555555555555555555555555555EEEE	Premium + payout currency
Premium Treasury	0x66666666666666666666666666666666FFFF	Receives premium payments
Underwriter Safe	0x77777777777777777777777777777777AAAA	Holds UNDERWRITER_ROLE
Oracle signer key	0x88888888888888888888888888888888BBBB	Holds ORACLE_SIGNER_ROLE on OracleAdapter

The policy catalog — used identically across every document

Three self-service plans, matching the three prices given in the request:

TemplateId	Plan	CoverageAmount	PremiumAmountPerPeriod	PeriodSeconds	TermSeconds
1	Basic	2,000 ether (2,000 dUSD)	11 ether (\$11)	30 days	365 days
2	Standard	5,000 ether	22 ether (\$22)	30 days	365 days

2	Standard	5,000 ether (5,000 dUSD)	22 ether (\$22)	30 days	365 days
3	Premium	10,000 ether (10,000 dUSD)	33 ether (\$33)	30 days	365 days

All amounts are 18-decimal fixed point (ether in Solidity = 10^{18}), matching every token in this project. \$33 = 33 ether = 33000000000000000000 in raw uint256 wei-equivalent units — written out once here since every later document just uses 33 ether for readability.

Index

Step documents (mechanics, read once):

1. Step-1-Wallet-Auth.md — connecting a wallet, EIP-191 login
2. Step-2-Buy-Policy-Premium-Payment.md — subscribeToPolicy, EIP-20
3. Step-3-File-Claim.md — submitClaim, the validity checks, EIP-1967/1822 proxy mechanics
4. Step-4-Oracle-Verification.md — EIP-712 signing and on-chain verification
5. Step-5-Payout.md — payoutClaim, the token transfer
6. Step-6-Whats-Not-Touched.md — Vault, PriceOracle, DICSGovernanceToken, DICSGovernor, ClaimGasPaymaster, EIP-2612, EIP-4337 — where they fit and where they don't

Scenario documents (concrete value-traces):

1. Scenario-1-Basic-Approved.md — \$11 plan, small claim, approved
2. Scenario-2-Standard-Approved.md — \$22 plan, mid claim, approved
3. Scenario-3-Premium-Approved-221.md — \$33 plan, \$221 claim, approved (the request's own example)
4. Scenario-4-Premium-Rejected-By-Oracle.md — \$33 plan, claim rejected
5. Scenario-5-Premium-Lapsed-Blocked.md — unpaid premium blocks a claim entirely
6. Scenario-6-Coverage-Exceeded-Blocked.md — claim larger than coverage, blocked
7. Scenario-7-Underwriter-Manual-Reject.md — a human override instead of the oracle
8. Scenario-8-Replay-Attack-Blocked.md — a security attack attempt, and why it fails